Power-network disruption data construction

Dataset construction explanation

This is a **synthetic teaching dataset**, not operational power-system data. The files are `power_substations.csv` (144 substations) and `power_links.csv` (344 undirected links). `generate_datasets.py` is a separate, staff-facing reproducibility script; neither student notebook contains data-generation code. NumPy's `default_rng(2042)` fixes the random draw sequence.

Timeline and column definitions

The link topology, load ratio and equipment age are known when the initial disruption is recorded. One substation is already interrupted at that point. `interrupted_later` records a simulated interruption over the following 24 hours only for the other substations. A blank for the initially interrupted substation means *`ineligible`*, not *`no later interruption`*. There is no simulated dynamic cascade, protective control or electrical load flow.

File and column	Meaning
Substations: `substation_id`	Unique identifier S001 to S144; for joining, never a predictor.
Substations: `load_ratio`	Simulated pre-disruption load relative to a nominal limit, uniformly drawn from 0.42 to 0.94, rounded to three decimals.
Substations: `equipment_age_years`	Simulated equipment age, uniform integer 2 to 40 inclusive.
Substations: `initially_interrupted`	1 for central substation S066, 0 for all others.
Substations: `interrupted_later`	Later binary outcome drawn only for the 143 initially functioning substations; blank for S066. There are 39 later positives in this fixed release.
Links: `substation_a`, `substation_b`	The two endpoints of a pre-disruption physical link; direction and weight are not represented.

Graph and target rule

Create a 12 by 12 grid graph with four-neighbour links, then choose 80 of its 121 square cells without replacement. Add one of the two possible diagonals to each chosen cell, at random. This creates local cross-links and varied clustering values while keeping the graph small and connected. Save each undirected link once. Degree counts links touching a substation. Local clustering coefficient is the proportion of possible links **among its direct neighbours** that are present.

For generating the later target only, standardise each of the four arrays below over **all 144 substations**, using its population mean and standard deviation, after rounding `load_ratio`. The data generator uses these simulated log-odds for substation *`i`*:

```
-1.04 + 0.70z(load_ratio_i) + 0.52z(equipment_age_years_i) + 0.46z(degree_i) +
0.75z(local_clustering_i)
```

Convert log-odds to probability with the logistic function and independently draw a Bernoulli outcome for every substation except S066. Assign a blank later outcome to S066. This construction rule deliberately creates possible associations with network position; it does **not** imply that the first failure travels over links. Student models must fit their own scaling on training observations only.

Interpretation limits

The graph is stylised, its links have no impedance or capacity, and interruptions are independent draws conditional on a constructed probability. Added features may yield mixed held-out results in this one sample. No model score proves failure propagation or usefulness for real-world grid operation. The initial interruption sets the time at which the prediction is posed but has no causal role in the outcome-generation rule.